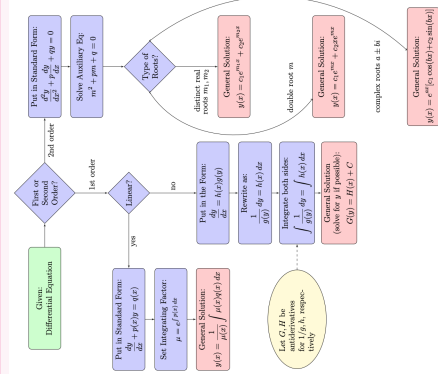


Theorem: Stokes' Theorem

Let  $\Sigma$  be a smooth oriented surface in  $\mathbb{R}^3$  with boundary  $\partial\Sigma \equiv \Gamma$ . If a vec field  $\mathbf{F}(x,y,z)$  is defined and has cont. first order partial derivs in a region containing  $\Sigma$ , then  $\iint_{\Sigma} (\nabla \times \mathbf{F}) \cdot d\mathbf{\Sigma} = \oint_{\partial\Sigma} \mathbf{F} \cdot d\mathbf{\Gamma}$ , or:  
$$\iint_{\Sigma} \left( \left( \frac{\partial F_z}{\partial y} dy dz \right) + \left( \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) dx dz + \left( \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) dx dy \right) = \oint_{\partial\Sigma} (F_x dx + F_y dy + F_z dz)$$

Remark: Solving A Differential Equation



Remark: Random Stuff

- $\nabla \cdot (\nabla \times \mathbf{F}) = 0$
- Gradient Thm: For  $C$  a curve from  $P \rightarrow Q$ ,  $\int_C \nabla F \cdot d\mathbf{s} = F(Q) - F(P)$
- $\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin(\frac{x}{a}) + C$  and  $\int \frac{dx}{\sqrt{a^2 + x^2}} = \frac{1}{a} \arctan(\frac{x}{a}) + C$
- $\nabla \times \langle f, g, h \rangle = \langle \frac{\partial h}{\partial y} - \frac{\partial g}{\partial z}, \frac{\partial f}{\partial z} - \frac{\partial h}{\partial x}, \frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \rangle$
- $\nabla(|\mathbf{x}|) = \frac{\mathbf{x}}{|\mathbf{x}|}, \nabla\left(\frac{k}{|\mathbf{x}|}\right) = \frac{-k\mathbf{x}}{|\mathbf{x}|^3}$

Definition: The Newtonian Universe

- An **event** is some  $(t, \mathbf{a}) \in \mathbb{R} \times \mathbb{R}^3$  labelling a fixed  $t$  in time and  $\mathbf{a}$  in space.
- Two events  $(t_0, \mathbf{a}), (t_1, \mathbf{b})$  are **simultaneous** iff  $t_0 = t_1$ . The distance between simultaneous events is the norm  $|\mathbf{a} - \mathbf{b}|$
- A **particle** is an object whose position in the universe is given at any time by one event
- A **worldline** is a particle's trajectory through the universe. It is a graph  $\mathbf{x}: \mathbb{R} \rightarrow \mathbb{R}^3, t \mapsto \mathbf{x}(t)$  of form  $(t, \mathbf{x}(t)) \in \mathbb{R} \times \mathbb{R}^3$ .
- The **configuration space** of a **many-body system** of  $n$  **particles** is the location of each particle in space, so is  $\mathbb{R}^3 \times \mathbb{R}^3 \times \dots \times \mathbb{R}^3 = \mathbb{R}^{3n} = \mathbb{R}^N$ .
- For the worldline of the  $a^{th}$  particle denoted  $\mathbf{x}_a(t): \mathbb{R} \rightarrow \mathbb{R}^3$ , the collection of the worldlines defines a curve  $\mathbf{x}(t) = (\mathbf{x}_1(t), \mathbf{x}_2(t), \dots, \mathbf{x}_n(t)): \mathbb{R} \rightarrow \mathbb{R}^N$  in the configuration space.

Definition: Newton's Equation

The PoD implies the equation  $\ddot{\mathbf{x}}(t) = \Phi(\mathbf{x}, \dot{\mathbf{x}}, t)$ ,  $\Phi: \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R} \rightarrow \mathbb{R}^N$ .  
For  $\mathbf{v}: \mathbb{R} \rightarrow \mathbb{R}^N$ , the system  $\begin{pmatrix} \dot{\mathbf{x}}(t) \\ \dot{\mathbf{v}}(t) \end{pmatrix} = \begin{pmatrix} \mathbf{v}(t) \\ \Phi(\mathbf{x}, \mathbf{v}, t) \end{pmatrix}$  has the same soln space.

Major Theorem: Newton's Laws

- A system of particles is **freely-moving** if  $\Phi(\mathbf{x}, \dot{\mathbf{x}}, t) = 0$ . Such a system with initial state  $\mathbf{x}(0) = \mathbf{x}_0, \dot{\mathbf{x}}(0) = \mathbf{v}_0$  is solved by  $\mathbf{x}(t) = \mathbf{x}_0 + t\mathbf{v}_0$ .
- A force is a map  $\mathbf{F}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  affecting a particle's velocity. For a particle of inertial mass  $m$ ,  $\Phi$  is a force field is  $\mathbf{F}(\mathbf{x}) = m\ddot{\mathbf{x}}$ . A point  $\mathbf{x} \in \mathbb{R}^3$  with  $\mathbf{F}(\mathbf{x}) = 0$  is an **equilibrium point** of  $\mathbf{F}$ .

Definition: Affine Transformation

Affine transformations preserve collinearity and ratios of distance

Definition: Galilean Transformations

A **Galilean transformation** is an affine transformation which preserves time intervals and the distance between simultaneous events

- A Galilean

transformation is given by a linear transformation of the form  $\begin{pmatrix} 1 & \mathbf{0} & s \\ \mathbf{v} & R & \mathbf{s} \\ 0 & 0 & 1 \end{pmatrix}$  where  $\mathbf{s}, \mathbf{v} \in \mathbb{R}^3, s \in \mathbb{R}$ , and  $R: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  is an orthogonal transformation.

$$\begin{pmatrix} 1 & 0 & s \\ 0 & 1 & \mathbf{s} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ \mathbf{x} \\ 1 \end{pmatrix} = \begin{pmatrix} t+s \\ \mathbf{x}+\mathbf{s} \\ 1 \end{pmatrix} \leftrightarrow \text{translation/change of origin in } \mathbb{R} \times \mathbb{R}^3, \quad (2.12)$$

$$\begin{pmatrix} 1 & 0 & 0 \\ \mathbf{v} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ \mathbf{x} \\ 1 \end{pmatrix} = \begin{pmatrix} t \\ \mathbf{x} + \mathbf{v}t \\ 1 \end{pmatrix} \leftrightarrow \text{motion with fixed, uniform velocity } \mathbf{v}, \quad (2.13)$$

and  $\begin{pmatrix} 1 & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ \mathbf{x} \\ 1 \end{pmatrix} = \begin{pmatrix} t \\ R\mathbf{x} \\ 1 \end{pmatrix} \leftrightarrow \text{re-orientation of spatial axes}, \quad (2.14)$

Theorem: Galilean Relativity

The laws of motion (Newton's Eq.) are invariant under Galilean transformations - translation, fixed  $\dot{\mathbf{x}}$  motion, and rotation of spatial axes.